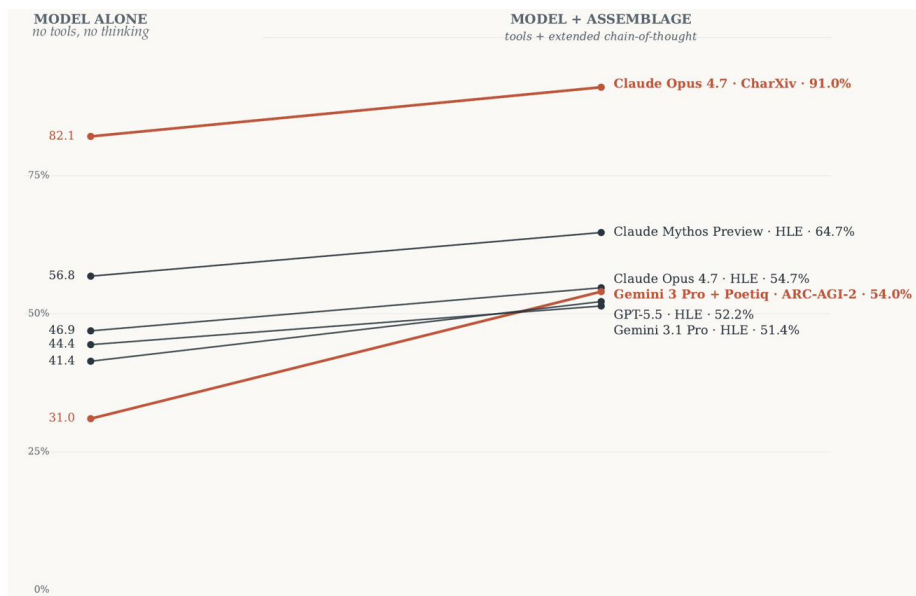


On the Non-Compositionality of Intelligence

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The University is just the way in which all that he has already seen is organized. When they are seen and when their coordination is understood, the University has been seen. (Gilbert Ryle, The Concept of Mind, 1949)



A quiet structural claim runs underneath much of what we say about artificial intelligence: that intelligence composes. Take a capable language model, attach an interpreter, a retriever, a filesystem, and an orchestration loop, and the result is intelligent in the same sense the model alone might be, only more so. The components are well-behaved, the composition well-engineered, the benchmarks reward the assemblage. We point at the assemblage and call it intelligent.

This essay argues that intelligence is **non-compositional under externalization**: the property we name when we say “intelligent” is not preserved when its constitutive operations are distributed across components outside the system whose intelligence we are claiming. A model that cannot compute without delegating is not a computing system; it is an interface to one. A model that cannot reason without delegating the reasoning is not a reasoning model; it is a router for one. This is not a claim about whether such systems

are useful: they are extremely useful. It is a claim about whether the words we use to describe them refer to anything specific. Most of the time, they don't.

The asymmetry shows up immediately. Ask a frontier model “what is 847×392 ?” and either it works through long multiplication in its scratchpad and arrives at 332,024 on its own, or it emits a tool call, an external interpreter returns 332,024, and the model relays the answer. From the outside both responses look like correct arithmetic. They are not the same kind of event. The first is computation. The second is delegation. Treating them as equivalent (as we routinely do when we report a model’s “math accuracy”) is the move this essay refuses.

The compositionality assumption

Some properties compose: mass, monetary value, deterministic correctness when each component is provably correct. Others don't. Formally, a property P of systems is *compositional* under a composition operator $\odot$ if $P(A) \wedge P(B) \Rightarrow P(A \odot B)$. The interesting question for intelligence is which $\odot$ we have in mind. Function composition $f \circ g$ preserves type-correctness; so does deterministic, infinite-precision arithmetic. Approximate, finite-precision, batch-fused floating-point arithmetic does not: associativity fails by $O(\epsilon \cdot n)$ in the worst case, where ϵ is unit roundoff and n is the reduction width.

Thinking Machines Lab’s analysis (He, Sep 2025) makes the structural case clearly. The popular story is that GPU floating-point non-associativity plus thread concurrency produces nondeterminism; He shows the actual culprit is **batch invariance failure**. Matmul, RMSNorm, and attention reduction kernels change numerical output for a given sample as a function of the surrounding batch, and batch composition shifts with concurrent load: *“Although the inference server itself can be claimed to be ‘deterministic,’ the story is different for an individual user. From the perspective of an individual user, the other concurrent users are effectively nondeterministic”* (He, 2025). Each component is run-to-run deterministic; the user sees nondeterminism. The published `batch_invariant_ops` library swaps RMSNorm, matmul, and attention via `torch.Library` and integrates with vLLM; one thousand temperature-zero runs collapse to a single bit-identical output, at a roughly 60% throughput cost. The earlier nondeterminism was real and located nowhere in particular.

The same seam reappears inside the training stack. The FlashRL group documented in 2025 that the trainer (HuggingFace forward pass) and the inference engine (vLLM) produce *different log-probabilities for the same sequence even with identical weights*, because their kernels differ — and quantization (FP8, INT8) widens the gap further. Concretely: let $\pi_{\text{train}}(\cdot \mid s)$ and $\pi_{\text{roll}}(\cdot \mid s)$

denote the trainer- and rollout-engine token distributions at state s . Even with identical parameters θ , the kernel differences yield $\pi_{\text{train}} \neq \pi_{\text{roll}}$, sometimes catastrophically: “for certain tokens, the two policies even yield contradictory predictions — $\pi_{\text{vllm}}(a, \theta) = 1$ and $\pi_{\text{fsdp}}(a, \theta) = 0$, which breaks the on-policy assumption and secretly makes the RL training become off-policy” (Yao et al., 2025). The standard on-policy PPO/GRPO surrogate is then biased; the only correction is a token-level truncated importance weight,

$$\mathcal{L}_{\text{TIS}}(\theta) = \mathbb{E}_{\tau \sim \pi_{\text{roll}}} \sum_t \min\left(\bar{\rho}, \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\text{roll}}(a_t | s_t)}\right) \cdot \hat{A}_t \cdot \log \pi_{\theta}(a_t | s_t), \quad (1)$$

with clip $\bar{\rho} \in [2, 5]$. ROLL Flash (Oct 2025) systematizes this family — TIS, IcePop, MIS, WTRS — as token-level and sequence-level corrections for the same underlying defect. The “efficient RL framework” is therefore implicitly off-policy; without (1) the gradient estimator silently corrupts. Each component is correct against its own specification. The bug exists only in the seam.

Deterministic components can compose into a nondeterministic system, because “deterministic” is not preserved under composition.

Intelligence fails to compose for analogous reasons. A language model is an excellent next-token predictor; an interpreter is a correct evaluator; a vector store is a faithful retriever. We compose them and call the result intelligent, but “intelligent” is not a property that composes the way we are asking it to. The understanding-and-solving is the assemblage’s behavior; crediting it to the model is the same category error as crediting determinism to a kernel that lacks batch invariance.

This inverts Ryle’s foreign visitor. The visitor commits a category mistake by treating the University as another entity alongside the colleges that constitute it. We commit the mirror error: shown the model, the tools, the retrieval layer, and the orchestration loop, we say “the model is intelligent.” Both are failures of categorization with the same origin: an intuition that the property we care about lives somewhere it does not.

When you say “the model can do X,” the question is what the model is. If it is the network weights, doing X means the weights compute X. If it includes the interpreter, the retrieval layer, the filesystem, and the agent loop, “the model” is shorthand for an engineering organization plus a deployment stack. Both senses are coherent in isolation; the problem is sliding between them. We report benchmarks under sense two and draw conclusions about capabilities under sense one. The agent’s contribution — selecting the tool, marshaling arguments, parsing results — is what a competent operator with a function-calling API does. We would not say the operator can do the math. We say the model can.

Four properties that fail to compose

Verification. A tool call is auditable; a native computation is opaque. Auditability composes; competence does not. A model that calls an interpreter correctly tells you nothing about whether the model could compute on its own.

Graceful degradation. A native system degrades smoothly; a delegating agent collapses sharply. By May 2026 the within-model gaps are smaller than at the start of the year, but they have not closed. On Humanity’s Last Exam (open-domain, contamination-resistant), GPT-5.5 scores 41.4% bare and 52.2% with tools; Opus 4.7 scores 46.9% / 54.7%; Gemini 3.1 Pro scores 44.4% / 51.4%; Mythos Preview, Anthropic’s withheld strongest model, gains 7.9 points (56.8% → 64.7%) when its full assemblage is brought online. CharXiv: Opus 4.7 jumps 82.1% → 91.0% with a single image-cropping tool. Anthropic’s own framing flags the no-tools number as isolating “the vision capability itself.” ARC-AGI-2: adding the open-source Poetiq harness raises Gemini 3 Pro from 31% to 54% at a 37× cost increase (\$0.81 → \$30.57/task). As of May 2026, no published frontier model scores higher without its tools than with them.

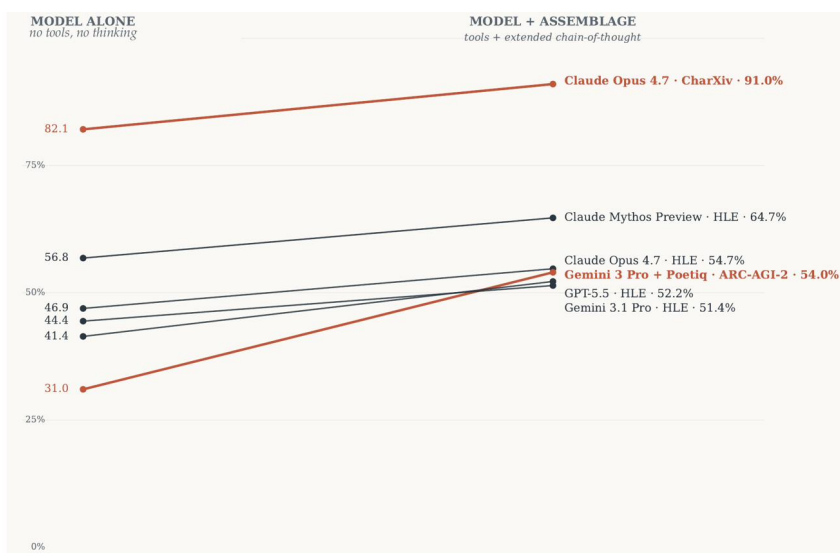


Fig. 1. The May 2026 externalization premium. Six model-benchmark pairs, each evaluated twice. The slope is what the assemblage adds. Anthropic flags that the Mythos HLE number “could indicate some level of memorization” at low effort. Sources: Opus 4.7 system card (Apr 16, 2026); Mythos Preview system card (Apr 7, 2026); GPT-5.5 launch (Apr 23, 2026); Gemini 3.1 Pro materials; ARC Prize Foundation Technical Report 2025.

OpenAI’s GPT-5 launch stated the disclaimer plainly: “AIME results with tools should not be compared directly to the performance of models without tool

access; they are an example of how effectively GPT-5 leverages available tools.” The same caveat appears verbatim on the o3/o4-mini launch page. The vendor knows the property does not compose. The boundary case is **ARC-AGI-3** (arXiv:2603.24621), released March 25, 2026, designed by François Chollet specifically to defeat extended chain-of-thought scaffolding. It drops the agent into 135 small interactive game environments with no instructions, no goals, and no rules disclosed. Crucially, evaluation runs offline on Kaggle: no external API calls, no cloud inference during scoring. Untrained humans solve every environment. At launch the four leading frontier models, run by their own vendors, scored a combined less than one percent.

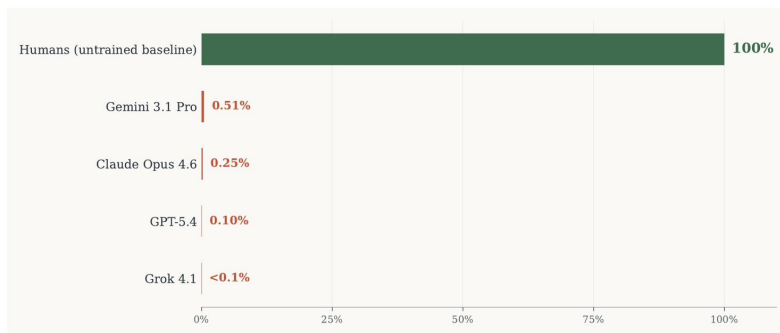


Fig. 2. When the assemblage cannot help. ARC-AGI-3 launch results, March 25, 2026. Both Opus 4.7 and GPT-5.5 shipped after the official evaluation; the public leaderboard remained below 1% across all submissions through April 30, 2026. When the benchmark defeats the very scaffolding that lifts scores everywhere else, the model alone returns nothing.

When the assemblage cannot help, the bare model has nothing to fall back on. This is the cleanest demonstration of non-compositionality the field has produced.

Composition with itself. A model computing natively can use its own outputs as inputs to its own further reasoning with the same numerical and semantic properties throughout. A delegating system round-trips through the tool layer, where every call is a serialization boundary. The training-stack analog is the trainer-versus-sampler numerics disagreement: when two components that should express the same model produce different log-probabilities for the same sequence, the gradient estimator is silently off-policy. The bug went undetected because each component was correct against its own specification.

Specification. When intelligence is internalized, the spec is the training distribution. When externalized, it is the union of the training distribution, every tool’s interface, every tool’s behavior, every parser, every retry policy, every error-handling branch, and every interaction between these. The latter

is enormously larger and almost none of it is documented. The system works because the team that built it has the spec in their heads and unit tests. None of this is the model. Calling the resulting artifact “intelligent” credits one component for the achievement of an entire engineering organization.

These four are projections of the same structural fact: intelligence as we use the word is a property of *where the work happens*, and externalizing the work moves the property out of the artifact we want to claim has it.

The brittle lesson of external intelligence!

The tool gap measures how much the assemblage adds; perturbation measures what the model itself has: what survives when the problem is shifted by something a competent reasoner would not notice. The 2025 literature is unusually clean and unusually unkind to the standard reading of benchmark scores.

ASyMOB (17,092 problems, 35,000 mathematically-equivalent variants) reports a frontier-average drop of 77.0% → 33.4%, with 43.6 points lost to substitutions a human would not register, with a maximum single-model drop of 70.3 pp. The most advanced models in the sweep (o4-mini, Gemini 2.5 Flash) score 96.8% / 97.6% on the seed and lose only 21.7 / 21.2 points. Frontier scale buys partial robustness, not full robustness. **Putnam-AXIOM** (variable/constant renaming) drops o1-preview from 41.95% to 22.35%, a 19.6-point absolute drop, 46.8% relative; ten of the remaining 18 models show non-overlapping 95% CIs. **MSCR** (single-word adversarial) reduces OpenAI o3 by 15.88 pp on GSM8K and GPT-4o by 17.21 pp via transferred substitutions; on open-source math models, one adversarial word can erase up to 49.89% of GSM8K accuracy. **RIDE** (item-response-theory difficulty conditioning) reports a −21.73 pp average across 26 models including GPT-5 and DeepSeek-V3.1.

The most controlled result comes from a different direction. Apple’s June 2025 study ran Claude 3.7 Sonnet (Thinking), DeepSeek-R1, and o3-mini high on Tower of Hanoi. Past a complexity threshold, accuracy collapses. Lawsen’s follow-up qualified part of this: some collapse past $N \approx 10$ is a token-budget artifact, and River Crossing includes mathematically impossible instances above $N = 5$. But the more arresting finding survived independent replication. As problems near the failure threshold, the models **reduce reasoning tokens despite ample budget remaining**, as if recognizing further effort will not help. The “Rethinking the Illusion of Thinking” replication (July 2025) confirmed: *“LRMs reach peak token consumption when the task is complex but still solvable. Conversely, when the model implicitly detects that the task lies beyond its capabilities,*

it allocates fewer resources.” The model gives up its budget before it fails. Not a reasoner pushing toward an answer it cannot reach, but a system that has learned to recognize the boundary of its own competence and back off, without internalizing the procedure that would solve the problem on the other side.

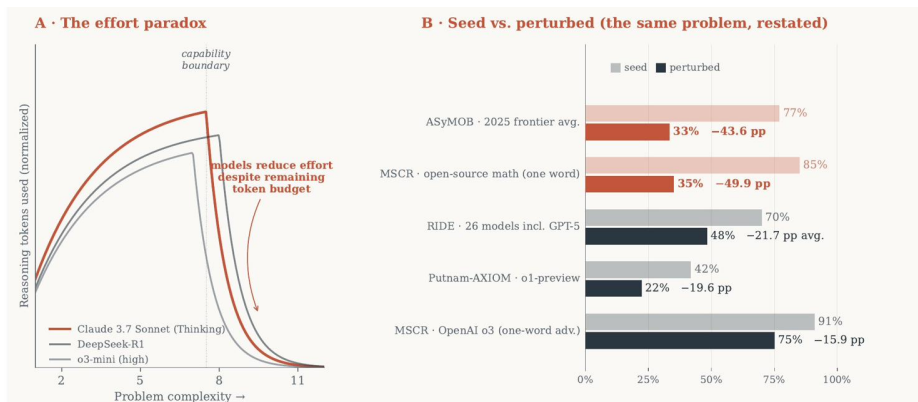


Fig. 3. What survives when the problem is shifted slightly. Panel A: three reasoning models on procedural tasks; reasoning-token usage rises with complexity then drops sharply despite remaining budget. Panel B: seed (lighter) vs. perturbed (full color) across five benchmarks; the two largest drops (MSCR open-source math, ASyMOB frontier average) highlighted.

What ties the tool gap and the perturbation collapse together is that they are projections of the same fact from opposite directions. The tool gap shows how much the assemblage adds; the perturbation collapse shows how little the bare model has. Either alone would be deniable. Together they leave the standard reading of benchmark scores indefensible: the model that scored ninety on the original cannot do thirty on a variant a competent reasoner would call identical, but it can do ninety on the variant if you let it write a Python function. The competence is somewhere. It is not where we have been pointing.

Where the work was, where it goes

A reading of the brittleness data: of course the bare model fails: capability had to be installed somehow, and “calling a tool” is just how installation manifests at inference. On this view, externalization at inference is the result of a training pipeline that itself externalized: a verifier in an RL loop, a teacher in distillation, a curated dataset for SFT. The complaint, on this reading, is just about timing.

The rejoinder has real force, because it points at the only mechanism by

which the externalization argument could be defused: gradient descent. If externalization gets internalized, the property eventually lives in weights, and the assemblage at inference is a contingent matter of cost. If it does not, externalization is structural and the attribution problem is permanent. Three families of recent results bear on whether and how internalization is happening.

The first is **RL with verifiable rewards (RLVR)**. Mukherjee et al. (2025) measured this directly across PPO, GRPO, ORPO, KTO, DPO, SimPO, PRIME, and ten model families: *“such large gains result from updating only a small subnetwork comprising just 5 percent to 30 percent of the parameters, with the rest effectively unchanged.”* RL induces parameter sparsity that emerges without sparsity-inducing regularization; the verifier’s signal is absorbed into a small, specific subnetwork; the rest of the model is left alone. Behaviorally: DeepSeek-R1 reported pure RL on a verifiable math reward took its base from 15.6% to 71.0% pass@1 on AIME 2024, 86.7% under majority voting, without SFT priming or inference scaffolding (the *Nature* update reports 77.9% for the pure-RL endpoint). By spring 2026 GPT-5.5 and Opus 4.7 reach near-saturation on AIME without tools. Mythos Preview reports 97.6% on USAMO 2026, an Olympiad-level set graded on proofs (which resist tool shortcuts) whose answers were not in any training corpus when scored. Most strikingly: on July 21, 2025, IMO President Prof. Dr. Gregor Dolinar officially confirmed an advanced version of Gemini Deep Think had earned 35/42 — gold-medal level — solving five of six IMO 2025 problems “in natural language within the 4.5-hour time limit,” no formal-language scaffold, no manual translation. A clean year-over-year improvement from AlphaProof, which had required Lean translation for IMO 2024 silver. The reasoning lives in weights.

The second is **distillation**. On-policy distillation (Lu / Thinking Machines Lab, Oct 2025) shows the student-on-its-own-rollouts variant *“can learn the RL-trained policy in approximately 7–10× fewer gradient steps, which corresponds to a compute efficiency of 50–100×”* — meaning verifier-distilled circuits can be transferred at roughly 1–2% of the RL cost. But the move is not free. Apple’s **Distillation Scaling Laws** (Busbridge et al., arXiv:2502.08606) report a sharp regime split: distillation only beats SFT when teachers already exist or when many students share one teacher; *“distillation does not improve upon standard fully supervised learning for LLMs given sufficient model size and compute budget on the student.”* And Kim et al. (arXiv:2603.24472, March 2026), *“Why Does Self-Distillation (Sometimes) Degrade the Reasoning Capability of LLMs?”*, document drops *“of up to 40%”* on out-of-distribution math when self-distillation suppresses epistemic verbalization — internalization can corrupt as well as compress.

The third is **mechanistic interpretability**. Anthropic’s circuit-tracing on

Claude 3.5 Haiku (March 2025) found two-digit addition is performed by two parallel pathways: one computing a rough approximation, the other determining the last digit precisely. The result is in weights at the circuit level. But the same paper documents an unsettling corollary: *“Claude seems to be unaware of the sophisticated ‘mental math’ strategies that it learned during training. If you ask how it figured out that $36+59$ is 95, it describes the standard algorithm involving carrying the 1.”* The verbal account is not faithful to the computation. This cuts both ways: it confirms internalization at the circuit level, but it also confirms that the chain-of-thought we point at as evidence of “reasoning” can be confabulation grafted over an unrelated internal computation. The post-hoc explanation is itself a form of externalization: produced for an external interpreter (the user) whose relationship to the model’s internal computation is contingent.

So the rejoinder has a sharply bounded form. **Externalization can be internalized: for bounded operations, by RL against a verifier, given enough compute and a clean reward signal.** This works for two-digit arithmetic, modular addition (Nanda et al. fully reverse-engineered the Fourier-circuit algorithm a transformer learns for $a + b \bmod p$), competition mathematics, and code under unit-test verification. It does not work for unbounded compositional reasoning, novel game environments, or distribution-shifted procedural execution. The brittleness data are exactly what it looks like when the verifier covered some operations and not others: operations the verifier saw are robust; operations one step outside its distribution collapse. The externalization argument is not a claim about an eternal limit; it is a claim about the present moment. Right now, in May 2026, the structural pattern holds: the work that constitutes the intelligence we are claiming happens in the toolchain at least as often as in weights.

There is one further reason the rejoinder cannot fully neutralize the argument, which has to do with bounded versus unbounded computation. The expressivity literature gives this an exact form. Let $\text{TF}_d^{\log}$ denote depth- d log-precision decoder-only transformers and let $\text{TF}_d^{\log}[t(n)]$ denote the same with $t(n)$ chain-of-thought decoding steps. Then under standard assumptions:

$$\begin{aligned} \text{TF}_d^{\log} &\subseteq \text{TC}^0, & \text{TF}_d^{\log}[O(\log n)] &\subseteq \text{TC}^0, \\ \text{TF}_d^{\log}[\Theta(n)] &= \text{L}, & \text{TF}_d^{\log}[\text{poly}(n)] &= \text{P}. \end{aligned} \tag{2}$$

A fixed-depth transformer without scratchpad cannot decide directed-graph connectivity, cannot simulate arbitrary automata, cannot solve linear equalities (Merrill & Sabharwal, ICLR 2024). With a linear number of intermediate generation steps it can simulate automata; with a polynomial number, Turing machines for that many steps. Length, not content, extends expressive power.

Even padding tokens with no informational content extend computational depth (Merrill & Sabharwal, NeurIPS 2025; Pfau, Merrill & Bowman, ICLR 2025). This is the formal version of “the bare model literally cannot do certain things; the scratchpad does computational work.”

But the depth can equally well live inside the model via recurrent depth iteration. Geiping et al.’s recurrent-depth model (NeurIPS 2025) shows a 3.5B-parameter network “*can improve its performance on reasoning benchmarks, sometimes dramatically, up to a computation load equivalent to 50 billion parameters*”, by iterating its hidden state at test time. Internal depth and external scratchpad are theoretically interchangeable in the relevant sense. The question is not whether intelligence requires depth (it does) but whether the depth lives *inside the artifact whose intelligence we are claiming*. The TC^0 argument supports internalization-as-possible; the brittleness data argue it is not yet what is happening at frontier scale.

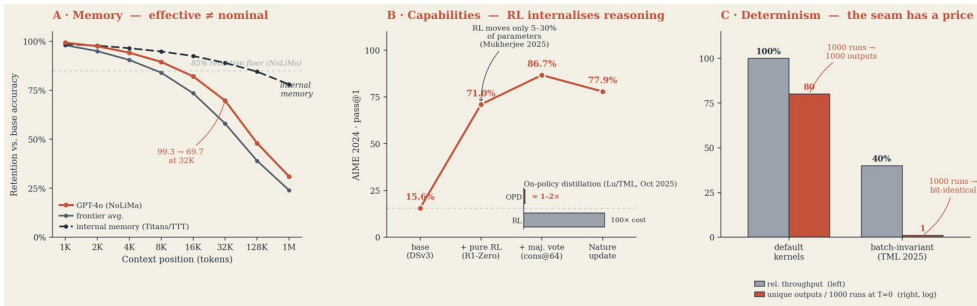


Fig. 4. Three axes along which the externalization line is currently being moved. A Memory: the gap between nominal and effective context length. NoLiMa (Modarressi et al., ICML 2025) reports GPT-4o falling from 99.3% to 69.7% by 32K tokens once literal-match cues are removed; frontier average degrades faster. Internal-memory architectures (Titans/TTT/Huginn) hold retention longer because the work happens in the activations, not in the prompt. **B Capabilities:** DeepSeek-R1’s pure-RL trajectory on AIME 2024 (15.6 → 71.0% pass@1, 86.7% with majority voting; Nature update 77.9%), with Mukherjee’s finding that RL across PPO/GRPO/DPO/etc. moves only 5–30% of parameters and the inset showing on-policy distillation reaching the same endpoint at ≈1–2% of RL compute (Lu/TML, Oct 2025). **C Determinism:** bit-identical inference is achievable but costs roughly 60% of throughput; the same kernel swap (TML 2025) collapses 1000 unique outputs at $T = 0$ to 1.

What “computing natively” actually requires

The right bar: the system’s latent state must be the locus of the relevant work. Not its outputs — outputs are cheap and any orchestration can produce correct outputs. The work has to happen in the activations. Whether with explicit

reasoning tokens, a hidden scratchpad, or pure latent-space reasoning is a second-order architectural choice. The first-order requirement is that the work happen somewhere inside the system whose intelligence we are claiming.

Not a counsel of perfection. No problem with using high-throughput external storage for unbounded knowledge (no model can fit the web in its weights). No problem with deferring genuinely irreducible operations (cryptographic verification, real-time sensor reads, communication with other agents) to dedicated systems. The objection is to systematic externalization of operations that constitute the intelligence we are claiming. Arithmetic is one. State maintenance over a long task is one. Stepwise reasoning under verification is one. If those leak into the toolchain, the model is not the thing being praised when the system performs well.

A research program recently active asks where capability has to live. On a casual reading it looks like architectural maximalism; the real move is austere. Build a system whose latent state simultaneously plays the role of compute, memory, and I/O. A maturity target is set by four conditions that must hold simultaneously: **Turing completeness**; **universal programmability** (capabilities installable by inputs and callable later); **behavioral consistency** under ordinary use; and **machine-native semantics** (primitives developed by the system rather than borrowed from a host).

Concretely: a system that solves any problem the human-plus-tools assemblage can solve, given equivalent compute, without externalizing the constitutive operations. A system you can program by handing it examples that it converts into circuitry it then calls. A system that produces the same answer at temperature zero across runs, replicas, and load conditions. A system whose explanations of itself are mechanistically faithful to its computations.

This is a high bar. Frontier production systems do not meet it: their compute is fixed-depth without scratchpad extension, their memory is short-context plus retrieval, their I/O is the tool layer. None of this is wrong as engineering. It is wrong as the answer to “is the model intelligent?” because it changes what “the model” refers to mid-sentence. The four-condition test is also what makes the externalization argument falsifiable: ship a system that passes it and behaves as well as the assemblage-based stacks, and the argument loses force.

An architecture sketch: the Closed-Loop Architecture

It helps to make the bar concrete. What follows is a single architecture, called the **Closed-Loop Architecture (CLA)**, that by construction satisfies the four-condition test without externalizing the constitutive operations. The point is

not that any production system today implements this; none does at frontier scale. The point is to show that the bar is specifiable. The pieces exist in the literature; they have not yet been composed into a single artifact, and frontier systems compose them with the tool-call layer instead.

Definition

Let $x_{1:T}$ be the input. The CLA is the tuple $(\mathcal{P}, \mathcal{R}, \mathcal{C}, \mathcal{M}, \rho)$ with:

1. Prelude $\mathcal{P} : \mathbb{R}^{|x| \times v} \rightarrow \mathbb{R}^{T \times d}$, a shallow encoder mapping tokens to per-position embeddings $e_{1:T} = \mathcal{P}(x_{1:T})$.
2. Recurrent block $\mathcal{R}_\theta : \mathbb{R}^{T \times d} \times \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^{T \times d}$, a parameter-shared block iterated $r(x)$ times (defined below). Inside $\mathcal{R}$, attention uses a non-diagonal, input-dependent state-transition update (§4) so that the per-token recurrence can recognize regular languages natively, not in token-space but in latent-space.
3. Coda $\mathcal{C} : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}^{T \times v}$, a shallow decoder.
4. Fast-weight memory $\mathcal{M}_t \in \mathbb{R}^{d_M \times d_M}$, an in-pass learned associative memory, updated by the rule (§5) below. $\mathcal{M}$ lives inside the forward pass; it is not a KV-cache offered to the user.
5. Recurrence policy $\rho : \mathbb{R}^d \rightarrow [1, R_{\max}(n)]$, a per-token learned policy choosing how many recurrence steps each position receives, with budget at least $R_{\max}(n) = \Omega(\log n)$ to clear the relevant complexity class (Merrill & Sabharwal, 2025).

The closed loop

The forward pass is the fixed point of an iterated map. With $s^{(0)} \sim \mathcal{N}(0, \sigma^2 I)$, for each position $i \in [T]$:

$$s_i^{(k+1)} = \mathcal{R}_\theta(e_i \oplus s_i^{(k)}; \mathcal{M}_t, S_t), \quad k = 0, 1, \dots, \rho(e_i) - 1, \quad (3)$$

where $\oplus$ denotes channel-wise concatenation, S_t is the recurrent attention state defined below, and the output is $\hat{y} = \mathcal{C}(s^{(\rho(e))})$. The loop closes inside the activation tensor: each iteration’s output is the next iteration’s input, with no token serialization in between. This is the architectural translation of “the work has to happen in the activations.”

Component 1: non-diagonal state transition (escaping TC^0 at the layer level)

Let $u_t \in \mathbb{R}^d$ be the input to a recurrent attention head at step t , and let $S_t \in \mathbb{R}^{d_k \times d_v}$ be its state. We use a generalized delta rule (RWKV-7 style) with input-dependent, non-diagonal transition:

$$S_t = (\text{diag}(w_t) - \hat{\kappa}_t \hat{\kappa}_t^\top a_t) S_{t-1} + v_t \hat{\kappa}_t^\top, \quad (4)$$

where $\hat{\kappa}_t = \kappa_t / \|\kappa_t\|$, and $(w_t, \kappa_t, v_t, a_t)$ are linear projections of u_t . The crucial property: the transition matrix $A_t = \text{diag}(w_t) - \hat{\kappa}_t \hat{\kappa}_t^\top a_t$ has eigenvalues outside the diagonal-only spectrum that confines diagonal SSMs (Mamba, S4) to TC^0 at finite precision (Merrill, Petty & Sabharwal, ICML 2024, “Illusion of State”). With (4) the per-token recurrence provably recognizes all regular languages natively, without buying expressivity from a polynomial-length scratch in the token stream.

Component 2: fast-weight memory (closing the long-tail loop)

The memory $\mathcal{M}_t$ is itself a small parametric model whose weights are updated within the forward pass by a Titans-style surprise-gated rule. Let (k_t, v_t) be projections of the recurrent state. Define “surprise” as the gradient of an associative reconstruction loss with respect to the current memory:

$$m_t = \beta_t m_{t-1} - \nabla_{\mathcal{M}} \ell(\mathcal{M}_{t-1}; k_t, v_t), \quad \mathcal{M}_t = (1 - \alpha_t) \mathcal{M}_{t-1} + \eta_t m_t, \quad (5)$$

with input-dependent gates: α_t (forgetting), β_t (momentum), η_t (learning rate). The associative loss is $\ell(\mathcal{M}; k, v) = \frac{1}{2} \|\mathcal{M}k - v\|^2$ in the linear case, or a deeper MLP variant. The update happens at each token, inside the forward pass, with no KV-cache exposed to the orchestrator. “Memory” here means: a small network whose weights change with the input the network is currently processing.

Component 3: adaptive recurrence depth (work where work is needed)

Tokens that demand multi-step composition get more iterations; tokens that don’t, don’t. The Mixture-of-Recursions policy ρ is trained jointly, with a budget regularizer:

$$\mathcal{L}_\rho = \mathcal{L}_{\text{task}} + \lambda \cdot \mathbb{E}_{i,x} [\rho(e_i)], \quad (6)$$

which selects the smallest depth that suffices per token. This is the architectural form of “think harder when you need to,” implemented in the layer count rather than in token count.

Component 4: single-stack numerics (closing the seam)

Training and inference share one numerical pipeline: batch-invariant kernels for matmul / RMSNorm / attention reductions (Thinking Machines Lab, 2025), unified FP8 for both the trainer forward pass and the rollout engine, and identical attention-mask conventions on both sides. Concretely, this enforces $\pi_\theta = \pi_{\text{train}} = \pi_{\text{roll}}$ at the per-token level, eliminating the need for the truncated-IS correction (1). The system is on-policy by construction, not by patch.

Component 5: verifier-distilled training, verifier-free inference

RL with verifiable rewards (DeepSeek-R1-style) is used during training only; the verifier is removed from the inference path. Mukherjee et al. (2025) provide the empirical handle: across PPO/GRPO/DPO/etc., 5–30% of parameters move under RL, and the moving subnetwork is consistent across seeds. The verifier signal is therefore localizable: after training, what the verifier taught is circuitry, not call sites.

The four conditions, checked

Turing completeness. With $R_{\max}(n) = \Omega(n)$ and unbounded scratch in the latent state (the per-iteration $s^{(k)}$ tensor), the looped architecture simulates Turing machines for $\text{poly}(n)$ steps (Giannou et al., ICML 2023; Merrill & Sabharwal, NeurIPS 2025). The scratch is the system’s own activations, not a token sequence handed to a sandbox.

Universal programmability. The fast-weight memory $\mathcal{M}_t$ is provably equivalent to a step of online gradient descent on an associative regression problem (Sun et al., ICML 2025; von Oswald et al., ICML 2023, “in-context learning is gradient descent”). Programs handed in via context become, mechanically, weights the system writes to a small in-pass network and calls later. “Programmability” is achieved without a tool registry.

Behavioral consistency. Single-stack numerics enforce bit-identical inference at $T = 0$ and bit-identical π_θ between rollout and update. The system has a determinable behavior, not an emergent one.

Machine-native semantics. The verifier-distilled circuits live in the 5–30% subnetwork that RL moves; mechanistic interpretability tools (sparse autoencoders, attribution graphs) operate on these weights, not on a tool-call log. The semantics are the system’s, not the host’s.

What CLA does not do

It is not the web. Long-tail factual recall still wants high-throughput external storage; CLA keeps that; it just does not call the database the model. It is not a sandbox. Cryptographic verification, real-time sensor I/O, and inter-agent communication still go to systems designed for them. The objection is to systematic externalization of operations that constitute the intelligence we are claiming: arithmetic, state maintenance, stepwise reasoning, length-extrapolating composition. Those go inside.

On RL, Distillation, Interpretability and Internalization of capabilities!

Recent research is converging from two directions that need to be kept apart.

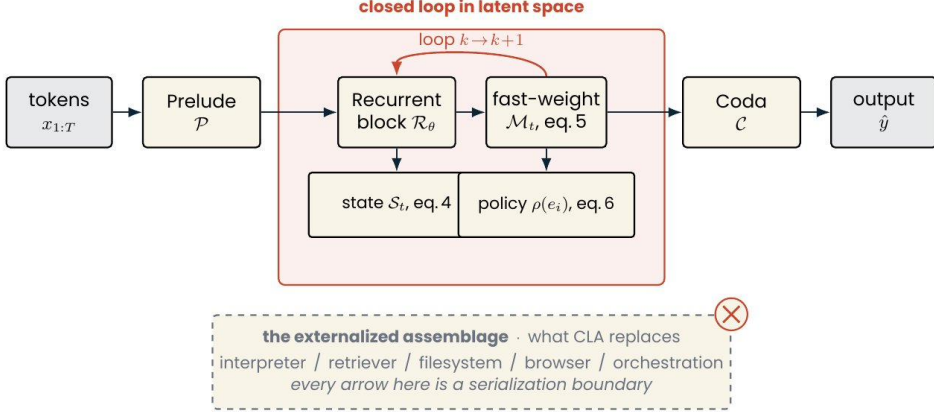


Fig. 5. The Closed-Loop Architecture. Five components: prelude $\mathcal{P}$, recurrent block $\mathcal{R}_\theta$ with non-diagonal state transition S_t , in-pass fast-weight memory $\mathcal{M}_t$, per-token depth policy ρ , and coda $\mathcal{C}$. The loop closes in the activation tensor: iteration $k \rightarrow k+1$ does not pass through the token stream. Below: the externalized assemblage that CLA replaces; every arrow that crosses out of the latent state is a serialization boundary.

On the internalization side of the line. Test-time-learning architectures collapse the controller-versus-memory distinction: Sun et al.’s TTT-layers (ICML 2025) make “the hidden state a machine learning model itself, and the update rule a step of self-supervised learning” — exactly (5) with linear $\mathcal{M}$ — and “can keep reducing perplexity by conditioning on more tokens, while Mamba cannot after 16k context.” Behrouz et al.’s Titans (Dec 2024) extends this to long-term neural memory with surprise-driven update gates and outperforms much larger models on the BABILong reasoning benchmark; the MIRAS follow-up (Behrouz, Razaviyayn & Mirrokni, 2025) recasts the design space as four components — architecture, attentional bias, retention gate, algorithm. RWKV-7 (Peng et al., March 2025) deploys (4) at 7B scale and proves regular-language recognition in Appendix D, the cleanest published architectural break of the TC^0 ceiling. Mixture-of-Recursions (Bae et al., NeurIPS 2025) implements (6) with per-token adaptive depth and reports Pareto-better perplexity at matched FLOPs with $2\times$ inference throughput. Geiping et al.’s Huginn-3.5B (NeurIPS 2025) trains the recurrent-depth template at 800B tokens and achieves “computation load equivalent to 50 billion parameters” by iterating its hidden state at test time. None of these systems composes all five CLA components simultaneously, and none has been pretrained at frontier scale; the architecture sketch is a specification of what such a composition would have to look like.

On the externalization side. The dominant production paradigm is what

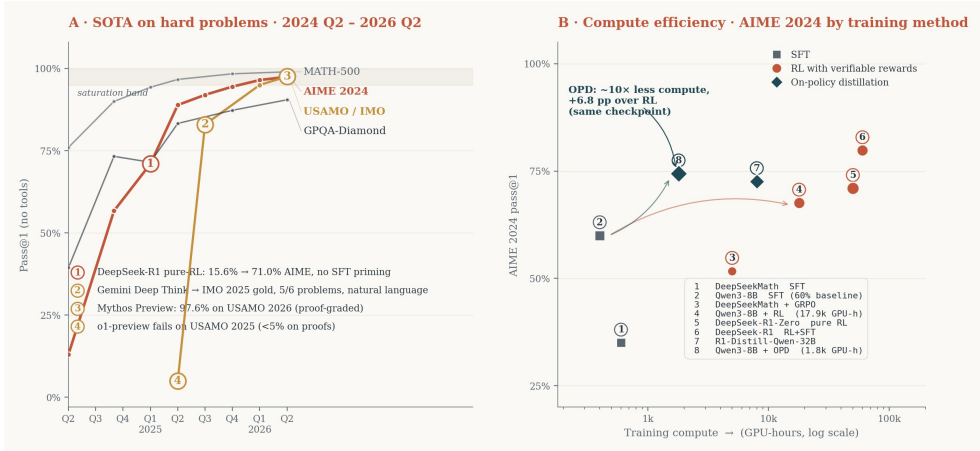


Fig. 6. Scaling on hard reasoning, 2024–2026. Panel A: AIME 2024 SOTA ascends through three post-training eras: test-time RL (o1-preview, Sep 2024), verifier-based RL (DeepSeek-R1, Jan 2025), on-policy distillation (Qwen3-8B, May 2025), and the frontier stack (Opus 4.7, Apr 2026). MATH-500 saturated by mid-2025; the same frontier models score below 1% on ARC-AGI-3 (Mar 2026). Panel B: Qwen3-8B same-checkpoint comparison. OPD reaches 74.4% AIME 2024 at 1,800 GPU-hours, strictly dominating RL’s 67.6% at 17,920 GPU-hours. Sources: model release scores from each lab; ARC Prize Foundation; Qwen3 Technical Report (May 2025); Thinking Machines Lab, On-Policy Distillation (Oct 2025).

Letta (the MemGPT successor) explicitly calls “LLM-as-an-Operating-System”: the LLM is like the CPU, and its context window is like the RAM, serving as the model’s working memory. Memory is external by design, accessed through tool calls (`memory_replace`, `memory_insert`, `archival_memory_search`). The benchmark situation is illustrative: Mem0 (Chhikara et al., arXiv:2504.19413) reports a 26% relative win over OpenAI on LoCoMo at < 7K tokens per retrieval; the original Zep paper claimed 84% on the same benchmark; Mem0’s reproduction corrected this to $58.44 \pm 0.20\%$; Zep’s counter-claim is 75.14%. Three numbers, one benchmark, the same harness ostensibly running. The point is not which framework is best — it is that a 25-point swing across vendor harnesses for the same memory operation is itself the externalization argument. Capability is not located at any one of the components.

The naming has caught up with the structure. By mid-2025 the field had renamed prompt engineering as **context engineering** — Karpathy, June 2025: “the delicate art and science of filling the context window with just the right information for the next step”; Tobi Lütke (Shopify), early summer 2025: “the art of providing all the context for the task to be plausibly solvable by the LLM”; Gartner, October 2025: “context engineering is in, and prompt engineering is out.” The renaming is itself an admission. We have stopped pretending the model is

doing the work alone.

The same gap shows up in long-context evaluation. **NoLiMa** (Modarressi et al., ICML 2025) defines “effective length” as the longest context where a model maintains $\geq 85\%$ of its base score and finds that *“even GPT-4o, one of the top-performing exceptions, experiences a reduction from an almost-perfect baseline of 99.3% to 69.7% [at 32K].”* At 32K tokens, eleven of the evaluated models drop below 50% of their short-context performance. Chroma’s **Context Rot** study (Hong et al., July 2025) extends this across 18 frontier models including GPT-4.1, Claude Opus 4, Gemini 2.5, and Qwen3: *“even under these minimal conditions, model performance degrades as input length increases, often in surprising and non-uniform ways.”* Effective context length is roughly 65–70% of nominal across the frontier as of mid-2026. The scaling curve in Fig. 4A is what this looks like as a structural fact: doubling the advertised window does not double the usable one, and the difference is exactly the externalization premium re-priced.

The latent-reasoning rebuttal has a sharper version worth pre-empting. Continuous-thought architectures (Coconut, Hao et al., Dec 2024) feed the model’s hidden state back as the next embedding rather than decoding to words, on the hypothesis that reasoning can occur in continuous representational space below the token level. Modest gains on small benchmarks. But a December 2025 causal/adversarial analysis (“Do Latent Tokens Think?”) found *“latent tokens in COCONUT showed minimal sensitivity to perturbations and displayed a clustered embedding pattern, further confirming that these tokens act as placeholders rather than meaningful representations of reasoning.”* The architecture is not yet a substitute for verbalized chain-of-thought at frontier scale: depth lives somewhere, but probably not in the latent tokens themselves, which is exactly the case for putting depth in (3) instead, where it is causally active by construction.

Verifiable-reward training applied beyond mathematics is the second front: theorem proving with proof checkers, code with unit tests, robotic manipulation with privileged-state physics simulators — domains where the verifier is an oracle at training and absent at inference. The DeepSeek-R1 trajectory (15.6 $\rightarrow$ 71.0% AIME via pure RL on a verifiable reward, with 5–30% of parameters moved per Mukherjee) is the proof of concept for the “verifier removed at inference” clause of CLA.

The honest cost

Frontier prototypes that try to internalize symbolic computation in a single architecture currently report something like four percent accuracy on two-

digit arithmetic, with reprompting (system-level conditioning) lifting this to the eighties, consistent with theoretical bounds on what a fixed-depth transformer can compute without scratchpad. But where the operation is reasoning rather than raw arithmetic, internalization has come further than pessimistic accounts allow. RL on a fixed pre-trained base with verifiable rewards and no learned reward proxy climbed from 15.6% to 71.0% pass@1 on AIME 2024, 86.7% with majority voting. The same paradigm with a proof checker reached IMO silver in 2024 and gold in 2025 with end-to-end natural-language reasoning. By spring 2026 GPT-5.5 and Opus 4.7 reach near-saturation on AIME without tools; Mythos hits 97.6% on USAMO 2026. None of these are orchestration wins.

CLA-style architectures pay in three currencies. The fast-weight update (5) requires a gradient step inside the forward pass; current TTT-Linear / TTT-MLP implementations run at $\sim 5\%$ of peak FLOPS on H100s versus $\sim 50\%$ for vanilla attention (“Test-Time Training Done Right,” arXiv:2505.23884). Recurrent depth (3) costs a multiplicative factor in latency proportional to $\mathbb{E}[\rho(e)]$. Single-stack numerics (Component 4) constrains kernel choice and forecloses some of the throughput optimizations production stacks rely on — concretely, the $\sim 60\%$ throughput cost of batch-invariant kernels in Fig. 4C. These are real costs. They are also costs of specifying what the system is, which is the sole charge of the four-condition test.

Externalization continues to win in production: it ships, it composes with existing infrastructure, it is what every frontier agent is built on. But the question is not “what works in production”; it is “what are we claiming when we talk about intelligence.” Conflating the two is the move I’m refusing. Imagine a chess engine that, asked for the best move, dispatches the position to a far stronger engine over an RPC and returns that engine’s answer. The wrapper is correct; by any benchmark it plays grandmaster-level chess. We would not, on this evidence, say the wrapper plays grandmaster-level chess. We would say it is good at calling the external engine. To know how good it is at chess, you would disable the tool layer and watch what happened.

The current state of frontier AI is, in many evaluations, closer to this analogy than is comfortable. Not in the trivial sense that models cannot function without tools, but in the deep sense that the reasoning, persistence, and calculation we point at when we say “intelligent” are happening, in the systems we deploy, on the far side of the tool boundary.

Generally intelligent systems and their internalization

A general intelligence is a system whose latent state is the locus of its computation, its memory, and its semantics — not a context window into which an engineering organization has loaded those things on its behalf. This is not a stylistic preference. It is a structural condition, and it is one the field has spent the last eighteen months independently arriving at from several different directions.

The clearest articulation is from outside this essay. In April 2026, Zhuge, Tian, Schmidhuber and colleagues (arXiv:2604.06425) proposed what they call a **Neural Computer** — “a machine form that unifies computation, memory, and I/O in a learned runtime state” — and gave a maturity test that is, line for line, the same four conditions this essay arrived at: Turing completeness, universal programmability, behavioral consistency unless explicitly reprogrammed, and machine-native semantics rather than borrowed-from-host primitives. They call the mature target a *Completely Neural Computer*; the essay calls it the Closed-Loop Architecture; the underlying claim is identical. The move under critique here has an older name in philosophy of mind. Adams and Aizawa, in *The Bounds of Cognition* (2008), called it the **coupling-constitution fallacy**: the slide from “X is causally coupled to Y during operation” to “X is part of Y, and Y has the cognitive properties of the whole.” To call an LLM-plus-orchestrator-plus-interpreter-plus-retriever a single intelligent system is to commit it. Two independent formulations of architecture converging on the same four conditions, with a thirty-year-old philosophical diagnosis sitting waiting for them, is evidence that the bar is not idiosyncratic. It is what *intelligent system*, as a term, has to mean if it is to mean anything.

What is genuinely new in 2025–2026 is that every CLA component now has a published, working instantiation; what does not yet exist is the composition. Recurrent-depth scaling: Geiping et al.’s Huginn-3.5B (arXiv:2502.05171, NeurIPS 2025 spotlight) trains a Prelude/Recurrent/Coda transformer at 800B tokens and shows that iterating the recurrent block 132 times at test time yields “computation load equivalent to 50 billion parameters,” with reasoning happening entirely inside the activation tensor rather than in emitted CoT tokens. Per-token adaptive depth: Bae et al.’s Mixture-of-Recursions (arXiv:2507.10524, NeurIPS 2025) implements the per-token policy ρ directly, with Pareto-better perplexity at matched FLOPs and $2\times$ throughput. Non-diagonal state transitions: Peng et al.’s RWKV-7 “Goose” (arXiv:2503.14456, March 2025) replaces the diagonal SSM transition with a generalized delta rule whose negative-eigenvalue spectrum provably exits the TC^0 ceiling Mamba and S4 are confined to (Merrill, Petty & Sabharwal, ICML 2024) — the paper proves NC^1 -completeness in a single layer and regular-language recognition in four. Fast-weight memory: Behrouz et al.’s ATLAS (arXiv:2505.23735, May

2025) extends Titans’ surprise-gated update with a generalized Omega rule and reports +80% on 10M-context BABILong, demonstrating that long-context retention through in-pass weight updates beats both attention and prior linear recurrences at the multi-million-token scale where they collapse. Behavioral consistency: Thinking Machines Lab’s batch-invariant kernels (He, Sep 2025), now upstreamed into vLLM and SGLang, collapse 1000 temperature-zero generations to 1000 bit-identical outputs at roughly 60% of throughput. Verifier-distilled training: Mukherjee et al. (arXiv:2505.11711) show across PPO, GRPO, ORPO, KTO, DPO, SimPO, and PRIME that RL fine-tuning consistently moves only 5–30% of parameters and that re-training the identified subnetwork alone reproduces the full effect — meaning the verifier-installed circuitry is localizable, separable, and lives in weights once the verifier is removed. The DeepSeek-R1 trajectory is the existence proof at scale: 15.6% → 77.9% pass@1 on AIME 2024, pure RL on a verifiable reward, no SFT priming, no inference scaffolding (*Nature* 645:633, Sept 2025). And on July 21, 2025, IMO President Gregor Dolinar formally confirmed Gemini Deep Think had earned 35/42 at IMO 2025 “in natural language within the 4.5-hour time limit,” no Lean, no AlphaProof translation, no external tools. A clean year-over-year improvement from AlphaProof’s 2024 silver. Mathematical reasoning at the limit of human competition is, at present, lived inside the weights.

The pieces are there. The composition is not. Sakana AI’s Continuous Thought Machine (Darlow et al., arXiv:2505.05522, NeurIPS 2025) is the most architecturally adventurous attempt at machine-native semantics: neuron-level temporal dynamics and neural synchronization as primitives the system develops itself, rather than borrowing token-by-token autoregression from the host language paradigm. It has not yet been demonstrated at frontier language scale. Nothing has composed all five CLA components at frontier scale. The architecture sketch in this essay is a specification of what such a composition would have to look like, and the empirical work has reached the point where that specification is no longer aspirational — it is what the field is, in pieces, already building.

An engineering analogue

The locality argument has a hardware version the chip-design community has been making for thirty years. Wulf and McKee’s 1995 note “Hitting the Memory Wall” observed that processor performance was improving ~60% per year while DRAM latency was improving ~7%; once the gap widens enough, system performance “is totally determined by memory speed.” Horowitz’s ISSCC 2014 plenary supplied the cost: on a 45 nm process, an 8-bit add costs ~0.03 pJ, a small SRAM read ~5 pJ, a DRAM access 640–2,600 pJ. Arithmetic is essentially free; data movement is not. Every tool call, every retrieval, every

vector-store read pays the Horowitz tax in a regime where the operations themselves are trivially cheap.

The hardware response has been a research program built on the observation that the dense linear algebra approximating intelligence in current architectures is non-compositional under externalization. Sebastian, Le Gallo, Khaddam-Aljameh and Eleftheriou (*Nat. Nanotechnol.* 15:529, 2020) framed the bet directly: data-centric AI workloads “call for a radical departure from the traditional systems.” IBM’s HERMES chip (Le Gallo et al., *Nat. Electron.* 6:680, 2023) is the existence proof — weights stored as phase-change-memory conductances, matrix-vector products performed by Kirchhoff and Ohm at 12.4 TOPS/W in 14 nm. NorthPole (Modha et al., *Science* 382:329, 2023) reports roughly $5\times$ the energy efficiency of an H100 on ResNet-50; the design principle, in Modha’s framing, “merges the boundaries between compute and memory.”

Fig. 7 plots the result. Each step closer to “compute and memory share a substrate” recovers an order of magnitude, and the cortex sits roughly $500\times$ above the best published silicon — not because biology has solved a hard search problem we have not, but because a single cortical pyramidal neuron is itself the input-output map of a seven-layer temporally convolutional network (Beniaguev, Segev & London, *Neuron* 109:2727, 2021): the closed loop closes inside the dendritic tree, with no scratchpad and no tool call. There is no synapse-level coprocessor. The non-compositionality argument this essay is making about LLMs is the argument hardware engineers have been making about von Neumann systems since 1995, with thirty years more empirical support and far less ambiguity about where the work happens.

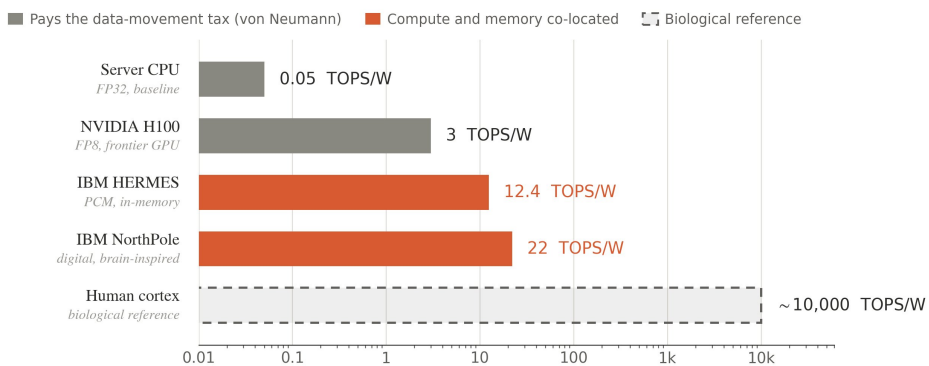


Fig. 7. *The locality premium. Inference efficiency on a common task (TOPS/W, log scale). The two coral bars share an architectural commitment the two grey bars do not: weights and arithmetic occupy the same physical substrate, so a matrix-vector product is a property of where charge sits, not of what travels across a bus. The cortex is not a target; it is the asymptote that fixes the slope. Sources: Horowitz, ISSCC 2014 (energy of arithmetic vs DRAM); Wulf & McKee, ACM SIGARCH 1995 (the memory wall); Le Gallo et al., *Nat. Electron.* 6:680, 2023 (HERMES, 12.4 TOPS/W in 14 nm); Modha et al., *Science* 382:329, 2023 (NorthPole, $\sim 5\times$ H100 on ResNet-50 inference); cortex an order-of-magnitude estimate from $\sim 10^{15}$ synaptic operations s^{-1} at ~ 20 W.*

The honest counterargument

The honest counterargument runs in the other direction, and it has to be engaged. Edwin Hutchins’ *Cognition in the Wild* (1995) shows in granular ethnographic detail that a navy ship’s bridge or an airliner cockpit *collectively* executes cognitive operations no individual member performs. “Speed bugs do not help pilots remember speeds; rather, they are part of the process by which the cockpit system remembers speeds.” The cognitive property is genuinely predicable of the distributed system. But Hutchins’s cockpit and an LLM-plus-tools stack differ in two structural ways. First, every component of the cockpit is itself a closed-loop system: pilots have brains, speed bugs are in continuous perceptual contact with airspeed indicators, the integrating mechanism is the pilot’s visual cortex comparing needle to bug at every cycle. Second, the integration is dense, recurrent, and in-substrate (visual coupling, manual control, voice loopback) rather than serialized through narrow textual function calls. Hutchins demonstrates that distribution is compatible with cognitive predication when the components are closed-loop and the coupling is constitutive. The LLM-plus-tools stack inherits neither property. The orchestrated assemblage is not Hutchins’ cockpit; it is the negative case the comparison is set up to refute.

The empirical counterweight to the internalization thesis is that orchestrated systems still win on raw benchmark capability today. Letta’s recent filesystem agent reaches 74% on LoCoMo by storing conversation history in a text file and giving the LLM `grep` — beating both Mem0 and Zep, the dedicated memory libraries. Sakana’s own AB-MCTS work (Inoue et al., arXiv:2503.04412) shows that no single frontier model exceeds 23% on ARC-AGI-2 under repeated sampling, but Multi-LLM AB-MCTS reaching across o4-mini, Gemini 2.5 Pro, and DeepSeek-R1 hits $\sim 30\%$; even the strongest published scaffolding (Gemini 3 Pro + Poetiq harness) reaches 54% only at a $37\times$ cost premium, while untrained humans solve every problem in the set. Test-time training layers, before LaCT’s chunked fix (arXiv:2505.23884), ran at under 5% of peak FLOPS against attention’s $\sim 50\%$. These are real numbers (Fig. 8). The position they support is not that internalization has won; it is that internalization is the path on which winning would mean something. AB-MCTS is doing externally what a future CLA system would do internally: parallel hypothesis exploration with adaptive branching. The architectural question is whether to keep that loop external — where each step is a serialization boundary, each model a black box to the others, each result an act of routing rather than reasoning — or to fold it into a substrate where the same exploration happens in activations.

A general intelligence is what you get when that fold succeeds. Not a model with tools. Not an orchestration loop over models. A system that,

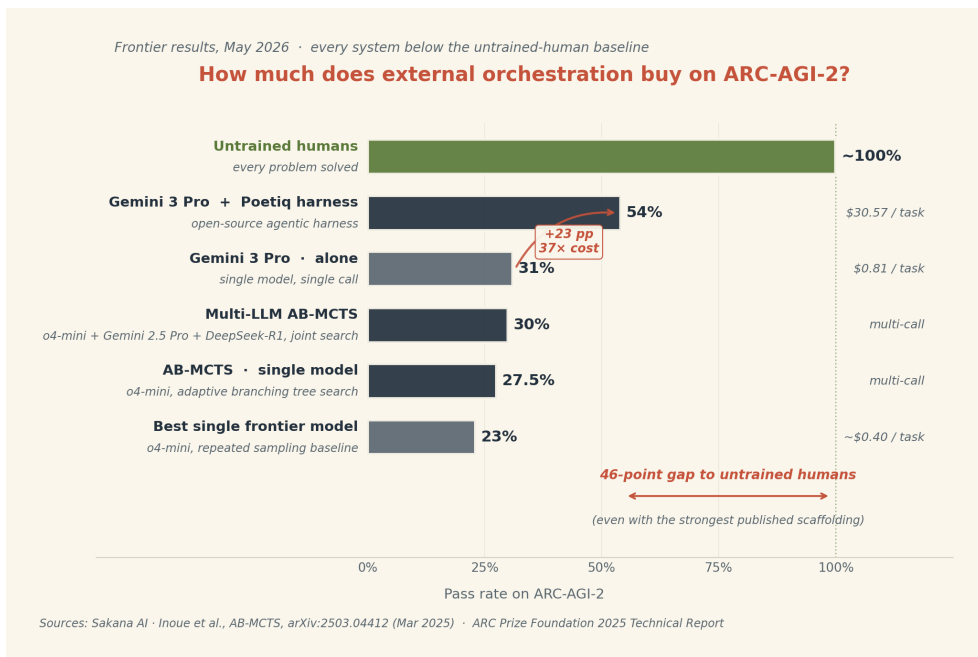


Fig. 8. The orchestration ladder on ARC-AGI-2. Frontier results, May 2026. Single-model repeated sampling on a strong base sits at ~23%; tree-search orchestration over that base reaches ~30%; the strongest published scaffolding (Gemini 3 Pro + Poetiq harness) reaches 54% at a 37× cost multiplier over single-call use (\$0.81 → \$30.57/task). Untrained humans solve every problem. Orchestration genuinely adds capability today, but at sharply rising cost and on a path that still leaves a humans-shaped hole; the internalization route is the one on which closing that gap would mean reasoning rather than routing better. Sources: Sakana AI / Inoue et al., AB-MCTS, arXiv:2503.04412 (March 2025); ARC Prize Foundation 2025 Technical Report.

given a problem, computes inside itself; that, given a piece of information, integrates it inside itself; that, given a procedure, installs it inside itself and calls it later. The four conditions are not aspirational targets. They are what the word *intelligent* refers to when it refers to anything specific. Without them, the word is doing the work of pointing at an engineering organization plus a deployment stack and saying “this thinks.”

A system that cannot compute without delegating to a filesystem is not a computing system; it is an interface to one. A model that cannot reason without delegating the reasoning is not a reasoning model; it is a router for one. The frontier today is full of useful, impressive, sometimes startling routers. We should be more careful about what we call them, because the systems that will actually be intelligent — in the strict, structural sense the four conditions name — are being built right now, in pieces, and we will need the word for them.

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[Editor's Note: Arguments are mine, writing is Claude's. This is partially an experiment in getting Claude to help speed-write and structure technical research blogs, drafted as artifacts and refined via "debate". I have too many blog ideas I never get around to writing up; the models finally feel good enough to help out (hopefully; let me know what you think).]